

Variables: T tiempo hasta la falla, r confiabilidad de un módulo, λ tasa constante, β, η parámetros de Weibull.

1. PROBABILIDAD

Discreta: $P(X = x) = p_X(x) \quad \sum_x p_X(x) = 1$

Continua: $P(a < X < b) = \int_a^b f_X(x) dx$

$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(u) du$

$f_X(x) = \frac{dF_X(x)}{dx}$

Esperanza: $E[X] = \sum x p_X(x)$

Varianza: $\text{Var}(X) = E[X^2] - E[X]^2$

Condicional: $P(A | B) = \frac{P(A \cap B)}{P(B)}$

Total: $P(A) = \sum_{i=1}^n P(A | B_i) P(B_i)$

Independencia: $P(A \cap B) = P(A)P(B)$

2. CONFIABILIDAD

Falla acumulada: $F(t) = P(T \leq t)$

Confiabilidad: $R(t) = P(T > t) = 1 - F(t)$

Densidad: $f(t) = \frac{dF(t)}{dt} = -\frac{dR(t)}{dt}$

Riesgo: $h(t) = \frac{f(t)}{R(t)} = -\frac{d}{dt} \ln R(t)$

Riesgo acumulado: $H(t) = \int_0^t h(u) du$

Relación: $R(t) = e^{-H(t)}$

MTTF: $E[T] = \int_0^\infty R(t) dt = \int_0^\infty t f(t) dt$

3. DISTRIBUCIONES DISCRETAS

Bernoulli: $P(X = x) = p^x(1-p)^{1-x} \quad E[X] = p$

Binomial: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad E[X] = np$

Geométrica: $P(X = k) = (1-p)^{k-1} p \quad E[X] = \frac{1}{p}$

Poisson: $P(X = k) = e^{-\mu} \frac{\mu^k}{k!} \quad E[X] = \mu$

Tasa constante: $\mu = \lambda t$

4. MODELOS CONTINUOS

Exponencial:

$f(t) = \lambda e^{-\lambda t} \quad F(t) = 1 - e^{-\lambda t}$
 $R(t) = e^{-\lambda t} \quad h(t) = \lambda \quad \text{MTTF} = \frac{1}{\lambda}$

Weibull:

$F(t) = 1 - e^{-(t/\eta)^\beta} \quad R(t) = e^{-(t/\eta)^\beta}$
 $f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^\beta}$
 $h(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1}$

5. SISTEMAS DE CONFIABILIDAD

Componentes independientes.

Serie: $R_S = \prod_{i=1}^n R_i$

Paralelo: $R_P = 1 - \prod_{i=1}^n (1 - R_i)$

k de n : $R_{k/n} = \sum_{j=k}^n \binom{n}{j} r^j (1-r)^{n-j}$

TMR (2 de 3): $R_{2/3} = \binom{3}{2} r^2 (1-r) + r^3 = 3r^2 - 2r^3$